"Cost benefit analysis is a good idea in principle, but is almost impossible to apply in practice to any important social decision". Discuss, with examples.

Core argument: In principle, Cost benefit Analysis (CBA) provides a coherent, Consequentialist framework for evaluating social welfare by measuring shadow prices, identifying Potential Pareto improvements, and rationalising public expenditure. However, when applied to major, non-routine social decisions - such as climate change mitigation, major infrastructure, health/safety regulations - CBA encounters severe practical and theoretical limits. These limits arise from:

1. Cognitive anomalies in non-market valuations
2. The 'tyranny of discounting' across inter-generational time horizons
3. Distributional biases and the arbitrary nature of the monetary numéraire
4. Record ~~bet~~ best distortions and large risks.

Consequently, while CBA remains a valuable screening tool for minor routine projects where market prices approximate shadow prices, it cannot serve as an objective or decisive engine for major social choice, where ethical choices, equity, and multi-criteria evaluation must be prioritised.

I would prefer to say "social welfare improvements" here since the P.P.I. criterion is problematic due to the issue of reversal paradoxes.

perhaps here you really mean "unweighted money-based CBA"?

Intro:

- introduce and explain principles of CBA ✓
↳ CBA is a systematic evaluation methodology designed to evaluate public sector decisions by quantifying their social gains and losses in monetary terms. — but the non-neutrality of this money numeraire will arguably then require "welfare-weighting"

- explain PPI criterion / KDH compensation
↳ public policy seen as socially advantageous if the aggregate ~~mone~~ monetary benefits accruing to the "winners" outweigh ~~the~~ the costs faced the "losers" ($\sum \Delta Y_i > 0$), such that the gainers could hypothetically compensate the ~~losers and t~~ losers and leave at least one individual better off without making anyone else worse off. ✓ As you discuss later on, the Kaldor–Scitovsky reversal paradox makes this problematic.

Paragraph 1: non-market valuation.

major social decisions routinely involve public goods, environmental assets, and human safety that lack observable market prices. ✓ To incorporate these into a Net present value calculation, economists rely on non-market valuation techniques. ✓

1. Revealed preference methods (travel cost models, hedonic property pricing). ✓
2. Stated preference methods; which construct hypothetical markets to elicit an individual's WTP or WTA ✓

Moreover, despite their wide spread use, stated preference models - particularly contingent valuation surveys - exhibit empirical anomalies which call into question ~~them~~ if survey responses reflect underlying economic utility.

1. explain embedding effect:
→ demand theory requires an individuals WTP for public good > WTP for small subset of that good.
→ in practice WTP = insensitive to scale/scope of good.

examples:
→ Bird conservation: mean WTP across 2000, 20,000, 200,000 birds dying was virtually identical.
→ Mine site clean up: mean WTP statistically indistinguishable between partial and full clean
→ wilderness protection: WTP to stop logging in 1, 2, or 3 areas was roughly the same.

2. explain 'warm glow' hypothesis
→ respondents use survey questions to express moral satisfaction or charitable support.
→ utility from the act of giving, not the specific amenity itself.
→ not suitable data for CBA calculations

Paragraph 2: tryanny of discounting

→ explain present value calculation:
$PV = \frac{B_t - C_t}{(1+s)^t}$ (benefit, cost), where s is the social discount rate

→ Contrast ~~social~~ social rate of time preference with marginal productivity of capital. ✓

$SRTP = \rho + \gamma \cdot g$ ✓

if return on capital > SRTP, individuals save and invest more. As capital accumulates, marginal return on capital falls until ✓ $r = SRTP$.

if time preference > capital return ✓, individuals consume more.

(This, however, is assuming no distortionary taxation!)

→ inter-generational problem: exponential discounting at standard commercial rates (e.g. 3-8%) nearly ~~plan~~ penalises future generations. ✓

↳ if A £1 billion catastrophic loss in the future is reduced to a present value of £7.6 million at a 5% discount rate. ✓ (100 years)

→ case study: climate change discounting rate.

Stern (2007): $\rho = 0.1\%$, justifies immediate spending

Nordhaus DICE: market-based discount rates where immediate abatement costs > distant climate change benefits

→ solutions? hyperbolic / time declining discount rates.
ecological discounting (seperating produced goods from environmental assets).

Paragraph 3: numéraire problem.

→ income bias of WTP: unweighted CBA operates on the implicit assumption that the social marginal utility of income is constant across all individuals ($\beta_i = 1$) (e.g. a pound is a pound)
↳ ~~CBA~~ WTP is constrained by income
∴ standard CBA biases choices towards preferences of high income groups.

→ Brekke (1997): selecting money as the baseline unit (numéraire) is not a neutral normalisation in economics unlike public goods.
→ switching the numéraire from monetary units to environmental units alters individual welfare weights by factors up to 1000×, and can flip social net benefit from positive to negative.

→ distributional weighting:
$dW = \sum \beta_h \cdot WTP_h$, where weights $(\beta_h = (M/m_h)^\epsilon$ reflect social inequality aversion
→ ~~not~~ hard to apply to real-world public policy, choosing weights requires ethical judgments + severe information constraints

Paragraph 4: 2nd best distortions

→ shadow pricing in distorted economies:
↳ in the absence of a first-best optimal state, market prices fail to reflect true social opportunity costs.
↳ (Brese + Stern): true shadow prices depend entirely on government policy responses, indirect taxation surpluses, Keynsian unemployment conditions and general equilibrium adjustments rather than simple market observations.

$v_i = \frac{\partial V^*}{\partial I_i}$, v_i shadow price: net impact on social welfare of a marginal increase in the supply of good i by the public sector.

→ in labour market with involuntary unemployment, hiring an unemployed worker does not imply a zero shadow wage. The shadow wage ~~must~~ ~~must~~ must account for the worker's reservation wage and the net social cost of income transfers.

→ for non traded goods, shadow prices must balance MSC with marginal consumer valuations (MSC_c), adjusted for tax revenue changes and distributional impacts

→ calculating these general equilibrium shadow prices requires extensive data on cross-price elasticities of demand, general equilibrium tax responses, and social welfare parameters. ✓

→ for major projects, such as the Channel Tunnel infrastructure development, the investment itself alters relative prices and market equilibria, violating the marginal assumptions of standard shadow pricing, requiring full structural general equilibrium modelling

→ Kaldor-Scitovsky paradox (Connolly, chpt 7)

↳ when a public project alters relative prices in ~~also~~ the economy, the utility possibility frontiers corresponding to the 'with project' and 'without project' states can intersect., ✓

↳ consequently, a project can pass a potential Pareto improvement test moving from State A to state B, while its cancellation simultaneously passes a potential Pareto improvement test moving from State B back to state A. ✓

Conclusion:

→ CBA's core objective - synthesising complex decisions into a single net present monetary value - is logically coherent in partial equilibrium environments

→ where CBA works: small, routine infrastructure investments (e.g. local road improvements, regional rail links) where market distortions are minor, timelines are short, and distributional impacts are negligible.
→ where CBA fails: major social choices involving non-market valuation, intergenerational time spans, ~~systemic~~ inequality, and severe scientific uncertainty (e.g. long term climate effects.)

→ CBA cannot serve as a single objective, decision making rule for important social decisions. It should be used in conjunction with cost-effective analysis, open democratic deliberation, and wider multi-criterion frameworks.

73 Overall this is a very good essay! You could also have mentioned how sensitivity analysis could be used to check the robustness of the results to the normative parameters chosen (e.g. the degree of inequity aversion and the social rate of time preference)